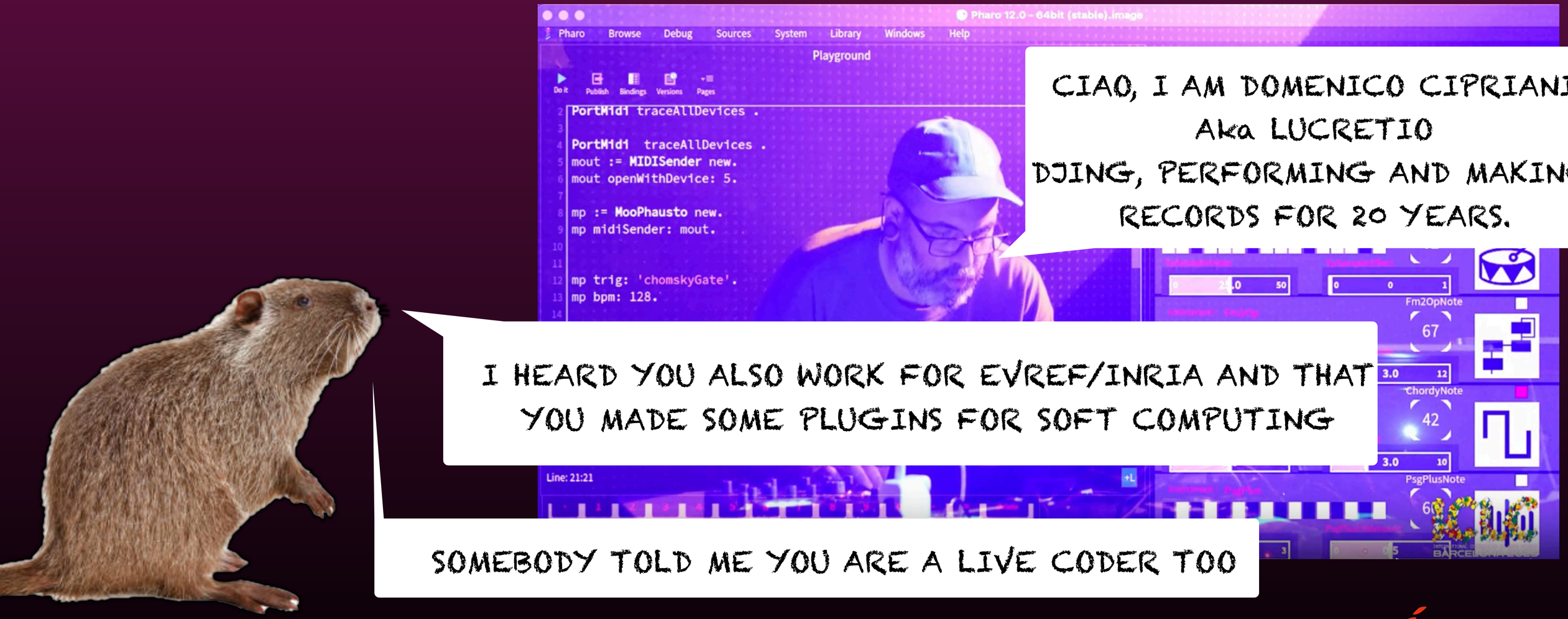
Accessible artful plugin design with Phausto and JUCE

Or, how to make audio plugins without touching C++

Domenico Cipriani - 2026

Inria

ABOUT ME



CIAO, I AM DOMENICO CIPRIANI
Aka LUCRETIO
DJING, PERFORMING AND MAKING
RECORDS FOR 20 YEARS.

I HEARD YOU ALSO WORK FOR EVREF/INRIA AND THAT
YOU MADE SOME PLUGINS FOR SOFT COMPUTING

SOMEBODY TOLD ME YOU ARE A LIVE CODER TOO

DISCLAIMER

Thanks for being here, thanks for being artful, thank for helping to make these tools better

I am very happy that there is still somebody that likes to write code for fun (and maybe work)



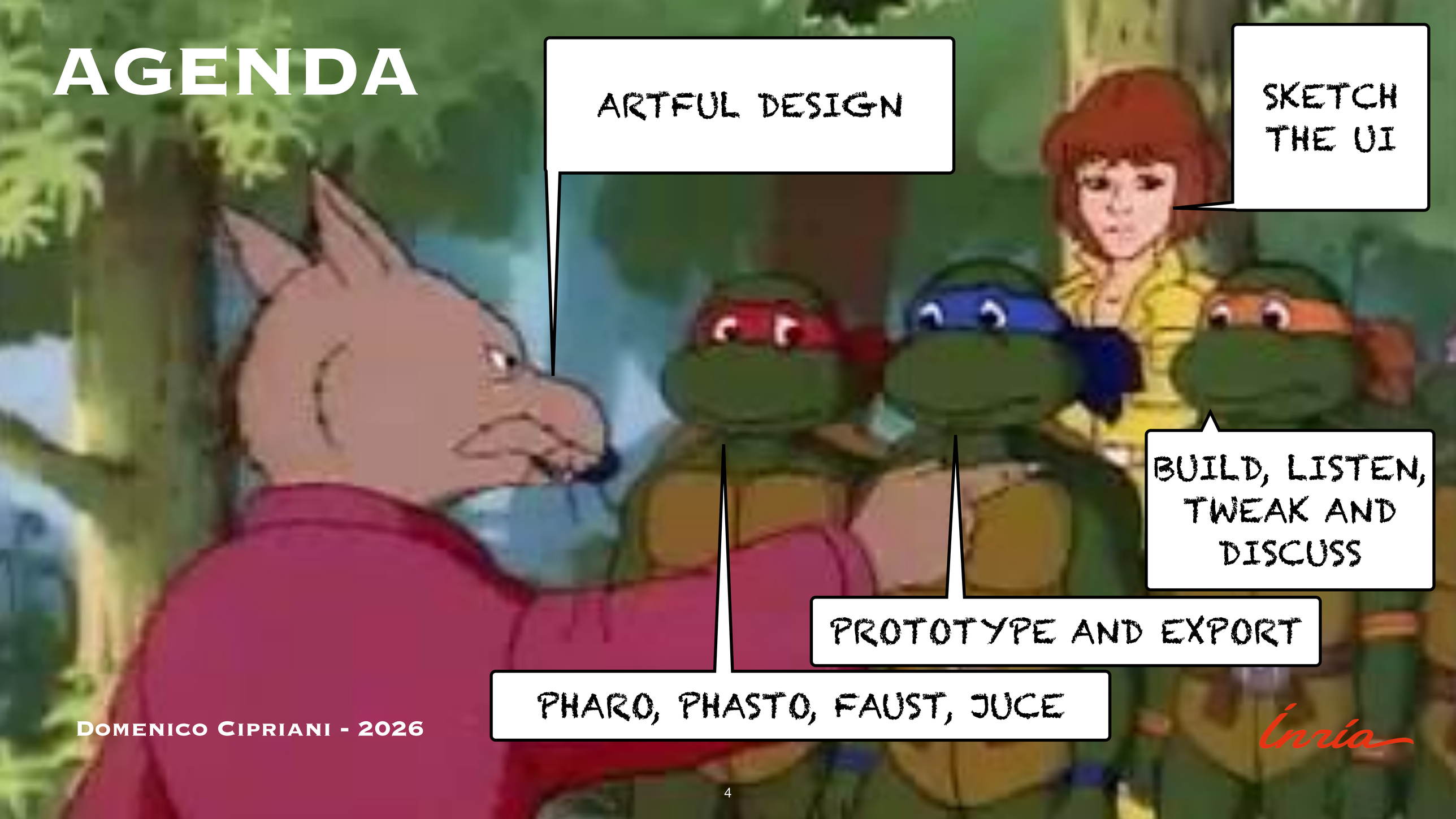
TECHNOLOGY SHOULD CREATE CALM



Domenico Cipriani - 2026

Inria

AGENDA


ARTFUL DESIGN
SKETCH THE UI
BUILD, LISTEN, TWEAK AND DISCUSS
PROTOTYPE AND EXPORT
PHARO, PHASTO, FAUST, JUCE



CAN A PLUGIN BE A PIECE OF ART?



BUT CAN WE DO IT WITHOUT C++?



YES WE CAN WITH PHARO AND PHAUSTO!

BUT SOONER OR LATER YOU SHOULD BETTER LEARN C++ TOO...





THAT'S ALL REALLY COOL,
BUT THAT WONT BE POSSIBLE WITHOUT JUCE!

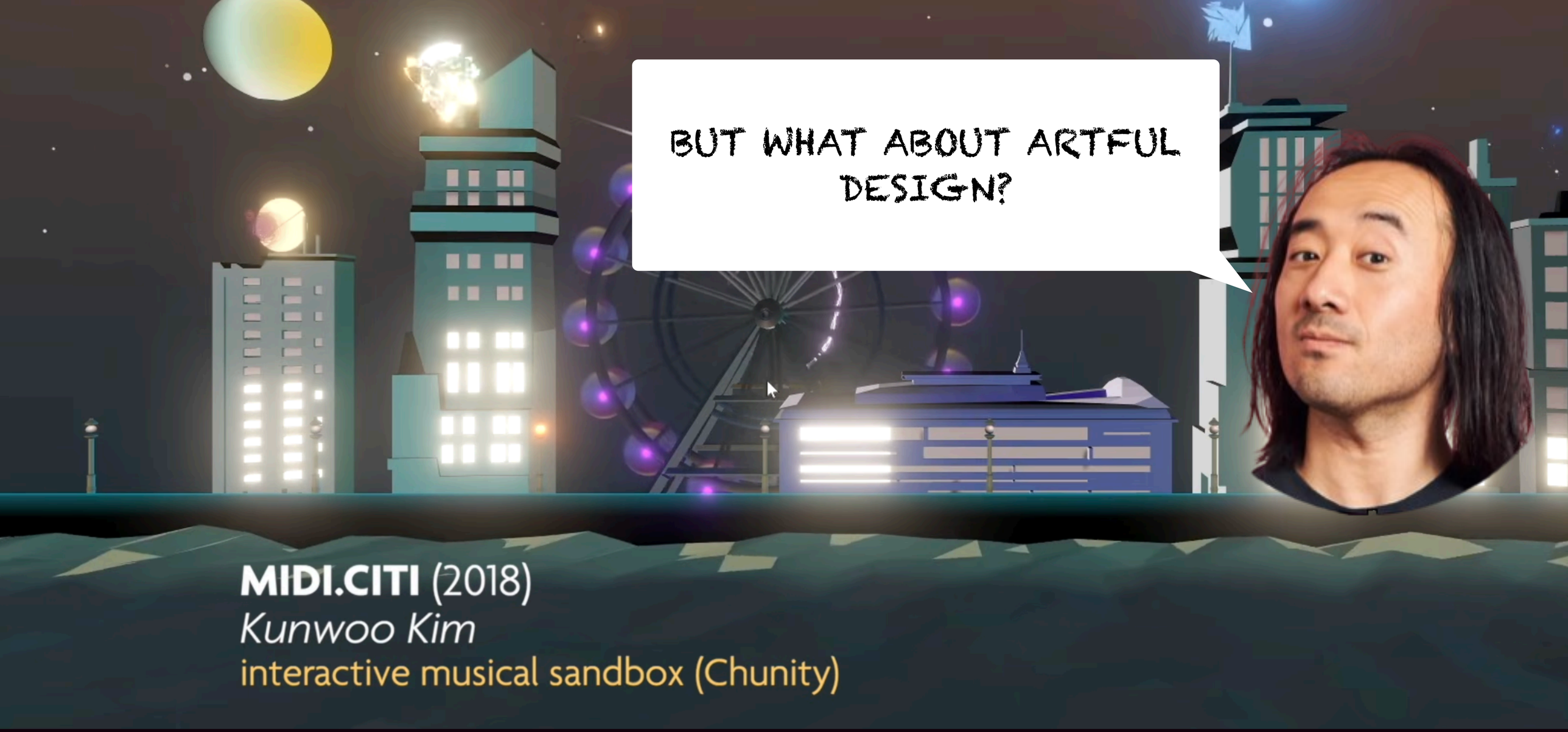
AND FAUST!!

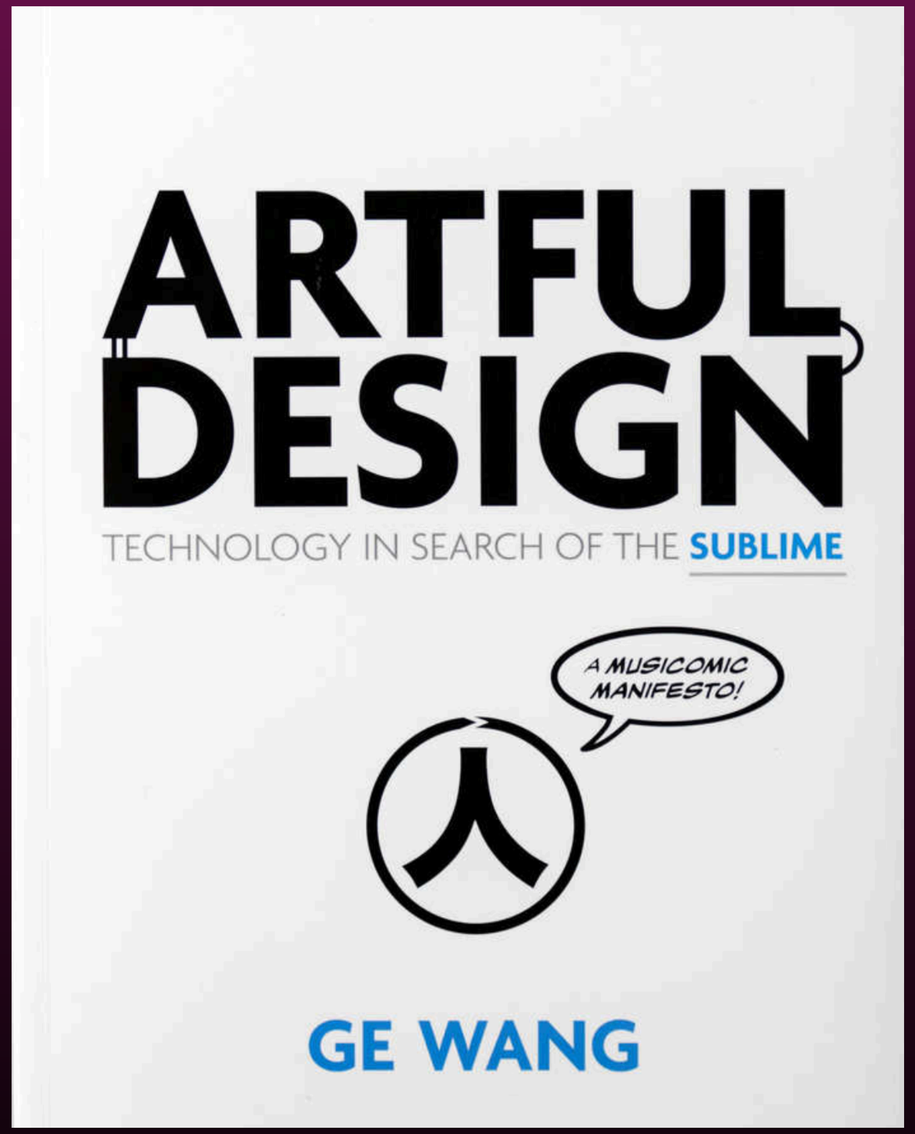


WHAT's YOUR PROBLEM WITH C++?

SLOW FEEDBACK LOOP,
HEAVY TOOLCHAIN,
REAL-TIME RULES,
SHARING ACROSS THREADS.
OWNERSHIP, LIFETIMES,
UNDEFINED BEHAVIOUR,
DEBUGGING BY EAR,
LIFECYCLE...
ENOUGH?????????????




BUT WHAT ABOUT ARTFUL DESIGN?
MIDI.CITI (2018)
Kunwoo Kim
interactive musical sandbox (Chunity)



In our age of evolving technology and human relentless and discord, Design ought to to be more thanks simply functional

It should be EXPRESSIVE, SOCIALLY MEANINGFUL and HUMANISTIC

Design should transcend the purely technological, encompass the human, strive for SUBLIME

PRAGMATICS
(how we live)
FUNCTION
NEEDS
WORK
MEANS-TO-ENDS

VS



AESTETHICS
(for what we live)
FORM
VALUES
PLAY
ENDS-IN-THEMSELVES

PHARO



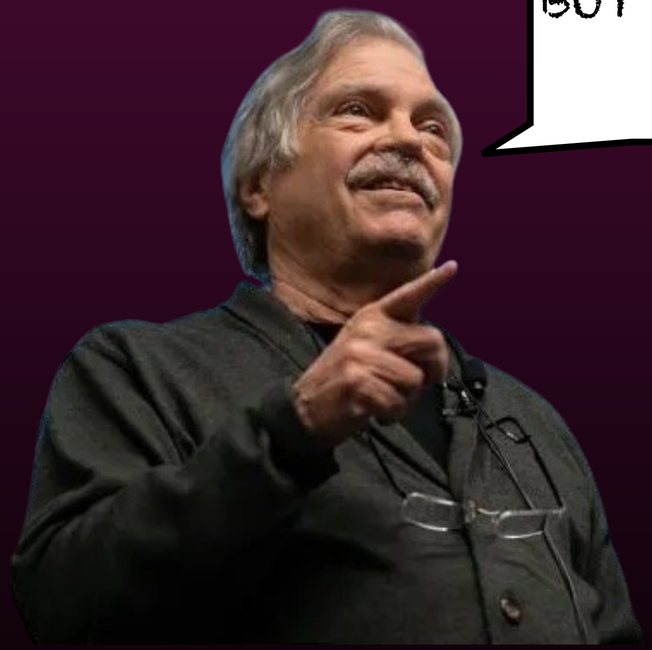
PURE OOP LANGUAGE + FULL IDE
INSPIRED BY SMALLTALK
FULLY INSPECTABLE AND REFLECTIVE

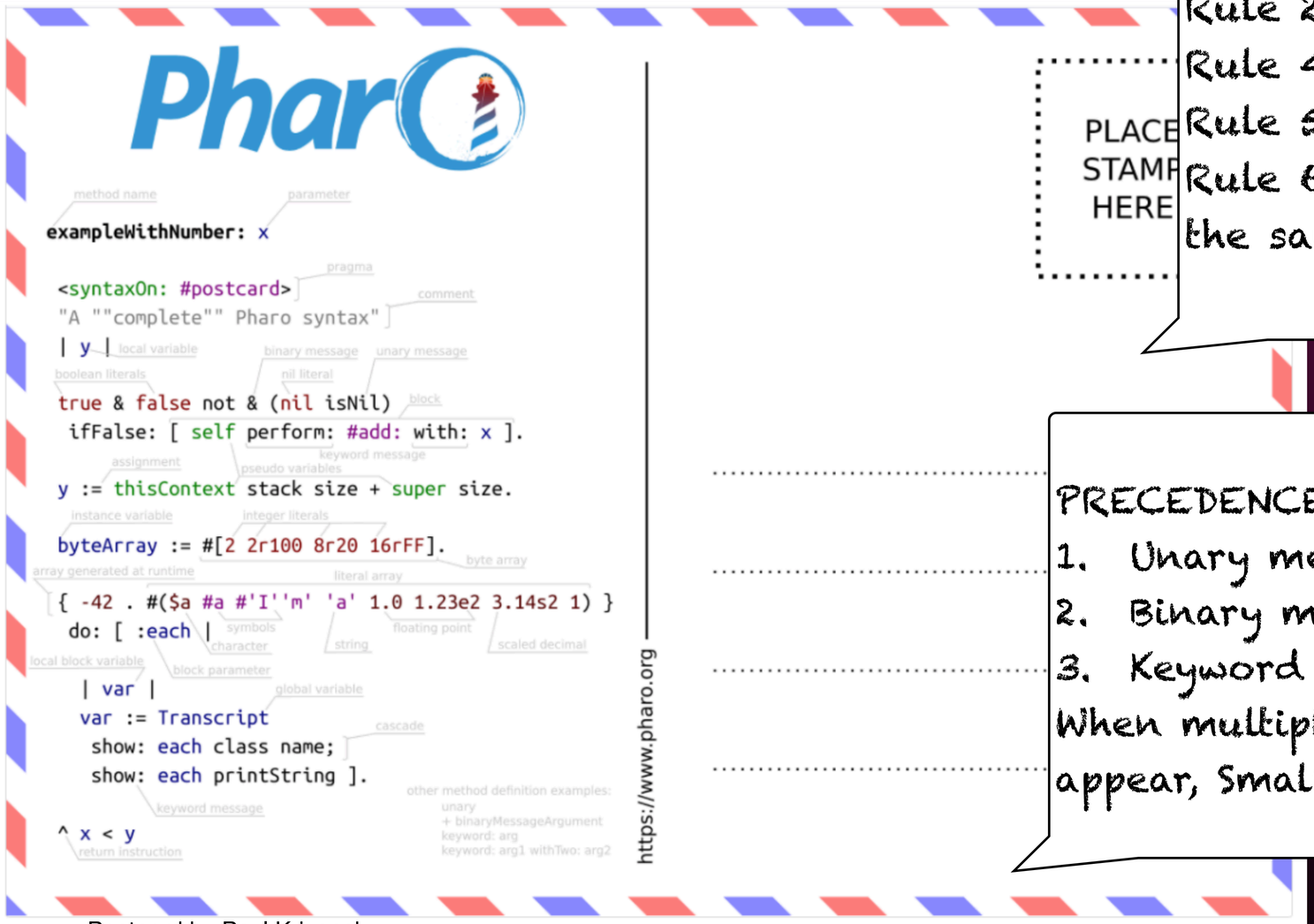
DYNAMICALLY TPED
ALL METHODS ARE PUBLIC VIRTUAL
ALL ATTRIBUTES ARE PROTECTED
SINGLE INHERITANCE

AND KYMA!



BUT DON'T FORGET TO TELL THEM ABOUT SMALLTALK!





Postcard by Paul Krivanek

PLACE STAMP HERE

Rule 1: Everything is an Object
Rule 2: Every Class has a superclass
Rule 4: Everything happens by sending messages
Rule 5: Method lookup follows inheritance chain
Rule 6 : Classes are Objects too and they follow the same rules

PRECEDENCE RULES:

1. Unary message (3 factorial)
2. Binary messages (3 + 5)
3. Keyword messages (Transcript show: 'Hello')

When multiple messages of the same precedence appear, Smalltalk evaluates them from left to right.

PHAUSTO

PHAUSTO IS AN API AND DSL FOR PROGRAMMING DSPs IN PHARO!



IT EMBEDS A FAUST COMPILER INTO A PHARO IMAGE AND INCLUDES MOST OF THE STANDARD LIBRARIES



```
1  osc1 :=  (PulseOsc  new duty: LFOSawPos new) freq: #freq.
2  osc2 := HsOscCos new freq: #freq.
3  env := AHDSRFEnvBias new trigger: #gate.
4  filter := OberheimLPF new normFreq: LFOTriPos new.
5  gain := #gain asPhHSlider .
6  voice := ((osc1 + osc2 ) * env * gain)  => filter .
7  dsp := (voice => ZitaRevStereo new)   asDsp.
8  dsp init.
9  dsp start.
10 dsp displayUI.
11 dsp stop.
```



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FAUST



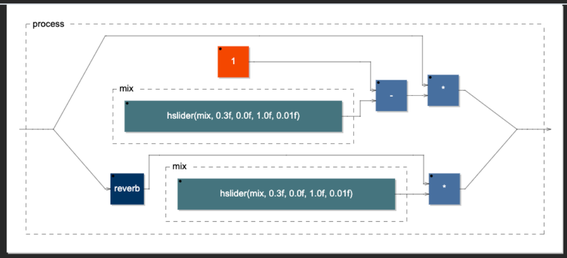
YOU KNOW FAUST, DON'T YOU ?

I AM THE BEST FOR DSP AND YOU CAN TRY ME IN THE BROWSER

Examples

```
1  declare name "PlainVerb";
2  declare description "A Schroeder reverb in plain Faust: no libraries, only
   primitives";
3
4  // Controls
5  size = hslider("size", 0.85, 0, 0.95, 0.01); // decay time
6  damp = hslider("damp", 0.4, 0, 0.9, 0.01);   // darkens the tail
7  mix  = hslider("mix", 0.3, 0, 1, 0.01);      // dry/wet
8
9  // One-pole lowpass: y = (1-d)*x + d*y'
10 lowpass(d) = *(1 - d) : + ~ *(d);
11
12 // Feedback comb: an echo every n samples, darker and quieter each time
13 comb(n) = (+ : @(n)) ~ (lowpass(damp) : *(size));
14
15 // Allpass: smears the echoes into a wash without colouring the sound
16 allpass(n, g) = (+ ~ (@(n - 1) : *(g))) <: @(n), *(-g) :> _;
```

Diagram | Plot | DSP



JUCE

I AM AT VERSION 9.0 , AND IAM OWNED BY THE PACE ANTI PIRACY GROUP

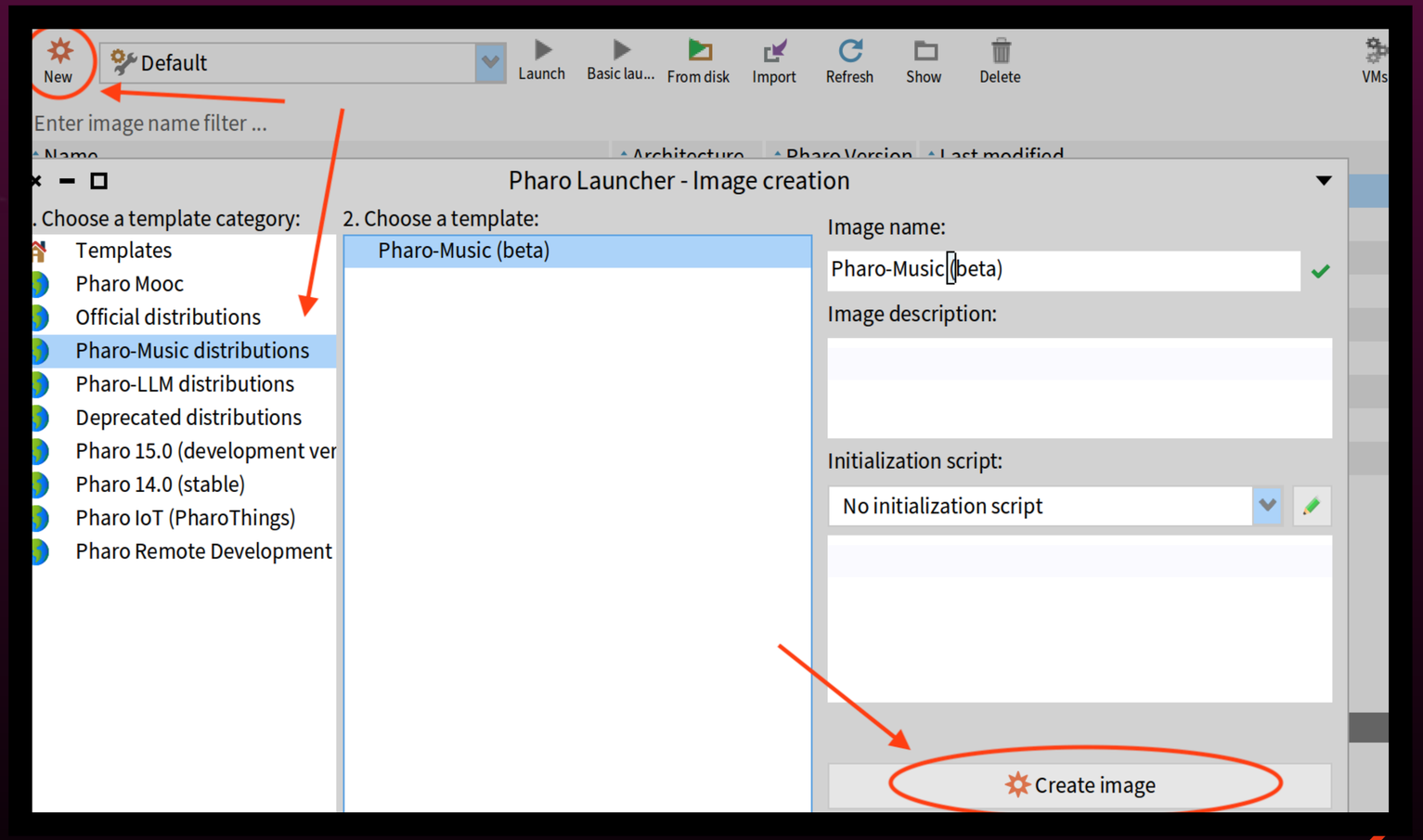


I CREATED JUCE IN 2004, ITS NAME STANDS FOR JUCE CLASS EXTENSION

IT IS AN OPEN-SOURCE C++ FRAMEWORK FOR BUILDING CROSS-PLATFORM STANDALONE APPS OR AS A VST/AU/AAX PLUG-IN ACROSS WINDOWS, MACOS, LINUX, IOS, AND ANDROID.
IT IS WIDELY CONSIDERED THE INDUSTRY STANDARD FOR AUDIO SOFTWARE DEVELOPMENT.

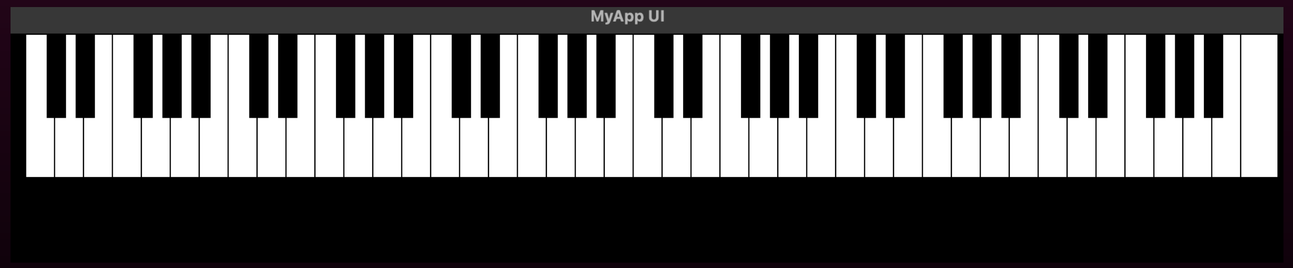
DOWNLOAD THE PHARO LAUNCHER AT HTTPS://PHARO.ORG/DOWNLOAD





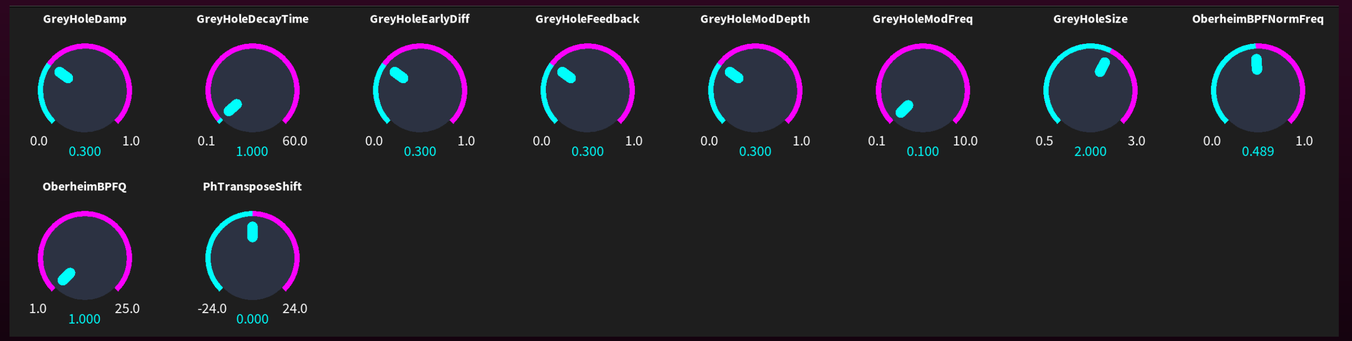
EXAMPLE1: A SYNTH

```
osci := SquareOsc new freq: #freq.
env := AREnv new trigger: #gate.
gain := PhHSlider new label: #gain.
dsp := ((osci => env) * gain) stereo asDsp.
dsp init.
dsp displayUI.
dsp export2JuceWithName: 'SqrSynth' type: PhMonoSynthPluginType new
```



EXAMPLE2: AN EFFECT

```
scubaFx := OberheimBPF new => PhTranspose new =>  GreyHole new.
dsp := scubaFx stereo asDsp.
dsp init.
dsp displayUI.
dsp export2JuceWithName: 'scubaFx' type: PhEffectPluginType new.
```




AND NOW YOU THINK I AM GOING TO USE AN LLM?